Cross-Species Modeling Identifies Gene Signatures in Type 2 Diabetes Mouse Models Predictive of Inflammatory and Estrogen Signaling Pathways Associated with Alzheimer's Disease

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Purpose of the Study

Type 2 Diabetes (T2D)
Inability to regulate blood sugar levels

Studies Have Reported
- Chronic Inflammation
- Increased Insulin Resistance
- Altered Glucose Metabolism in Brain

81% of people with AD have T2D or impaired fasting glucose

Mechanism ?

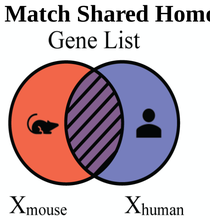
Risk Factors
Lifestyle, Sex, Genetics, Environment, Age → Alzheimer's Disease (AD)

Objective: To apply a cross-species modeling approach using publicly available transcriptomics mouse and human data to identify potential biological pathways associated with Alzheimer's disease.

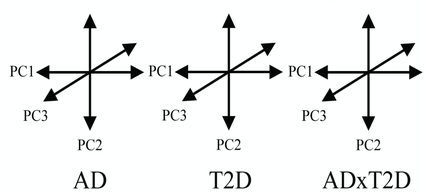
Translatable Components Regression (TransComp-R) Workflow

Step 1. Project Human Samples into Mouse Principal Component Analysis (PCA) Space

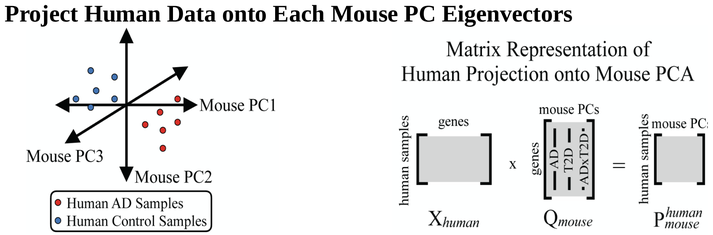
1. Match Shared Homologs



2. Construct Mouse PCA Spaces

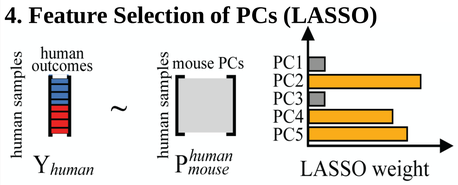


3. Project Human Data onto Each Mouse PC Eigenvectors



Step 2. Determine Principal Component (PC) Translatability

4. Feature Selection of PCs (LASSO)

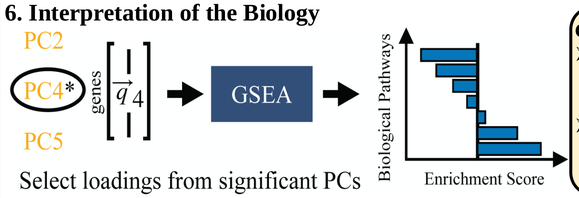


5. Linear Regression of Selected PCs

$$Y_{human} = \beta_0 + \sum_{i=1}^{k} \beta_i \, (X_{human} \vec{q}_i)$$

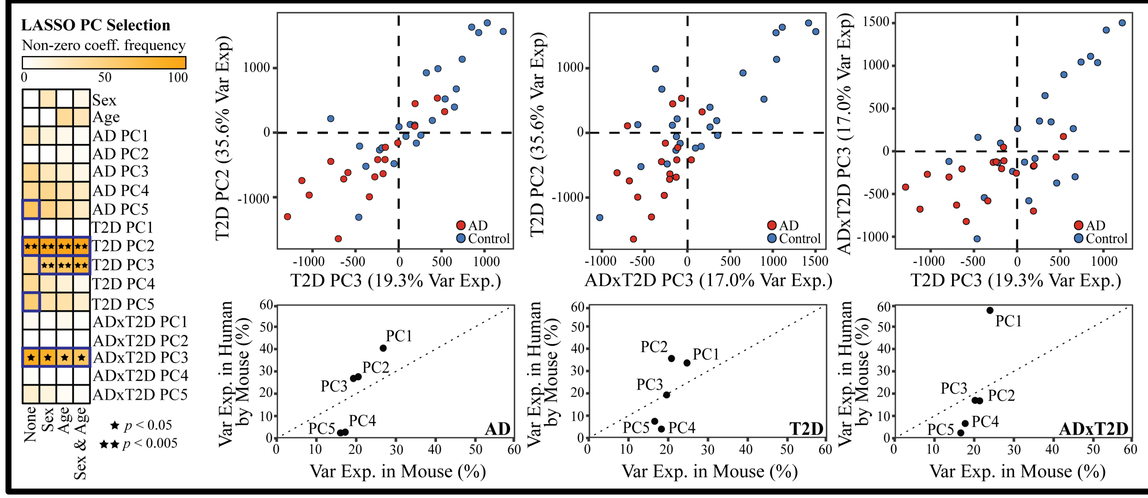
$$Y_{human} = \beta_0 + PC2 + PC4 + PC5$$

6. Interpretation of the Biology



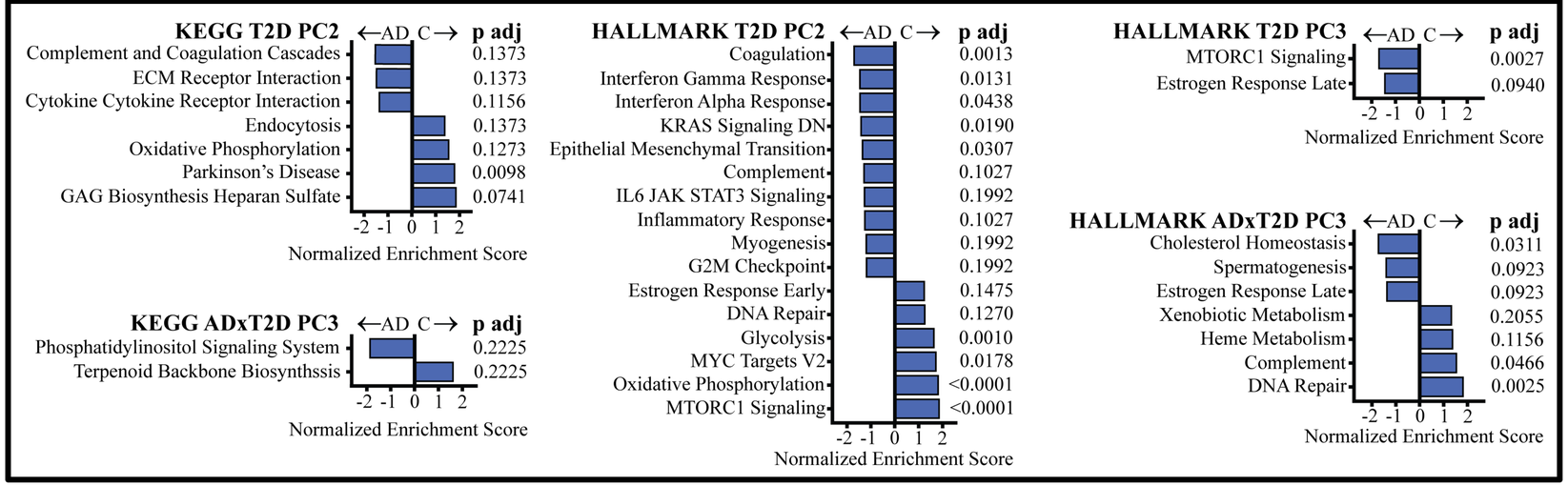
Gene Set Enrichment Analysis
- GSEA provides potential biological pathways by inputting lists of gene expression
- This provides a way to interpret results from TransComp-R

TransComp-R Modeling Separates Human Samples into Mouse Principal Component Space

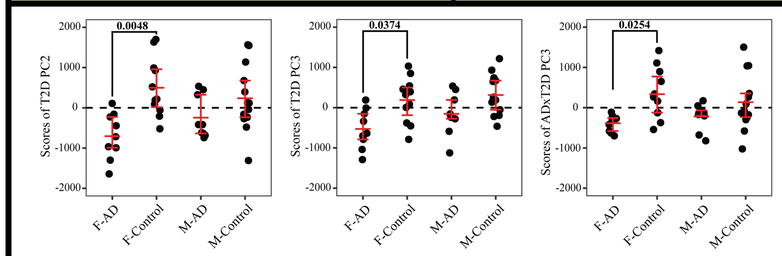


Key Findings:
- Four separate LASSO models, each ran across 100 rounds of 5-fold cross-validation, consistently selected mouse PCs associated with T2D and ADxT2D mouse models
- Mouse transcriptomic variation encoded on T2D PC2, T2D PC3, and ADxT2D PC3 best separated human AD and control groups
- T2D PC3 and ADxT2D PC3 explained similar variance in ratio, whereas T2D PC2 explained almost double in human compared to mouse data
- These PCs were next analyzed through gene set enrichment analysis

Gene Set Enrichment Analysis Identifies Inflammatory and Estrogen Signaling Pathways Enriched in Alzheimer's Disease



Male Derived Mouse PCs Stratify Female AD and Controls



Future Directions

In this work, we find biological pathways associated with inflammatory and estrogen signaling. Moving forward, further analysis of these pathways and validating these findings in human studies may help us develop potential diagnostics or therapeutics for Alzheimer's disease.